📘 Dashboard Update – Fleet Management System

🎯 Objective

Develop an integrated system to manage and track company vehicles, enabling:

* Asset management (vehicles)
* Vehicle assignment to employees
* Usage and operational cost tracking
* Accurate and printable reporting

1️⃣ Vehicles Inventory

Description:

Introduce a dedicated module within the dashboard to manage vehicles as independent assets.

Requirements:

* Create a “Fleet / Vehicles” section
* Ability to:

  * Add new vehicles
  * Edit vehicle details
  * Delete or archive vehicles
  * View vehicle list

Core Fields per Vehicle:

* Unique ID
* Vehicle Index Number
* (Optional – future): plate number, model, year, status

2️⃣ Vehicle Assignment System

Description:

A system to assign vehicles to employees with precise time tracking.

Requirements:

* Link each vehicle to an employee (technician or admin)
* Record:

  * Assignment start date & time
  * Assignment end date & time

Business Rules:

* A vehicle cannot be assigned to more than one employee during the same time period
* The current assignment must be closed before creating a new one for the same vehicle

Output:

* A chronological record of vehicle usage across employees

3️⃣ Activity Log

Description:

A complete audit trail for all vehicle-related activities.

Must Capture:

* User (employee)
* Activity type:

  * Assignment / handover
  * Usage
  * Notes
* Time period
* Activity details

Operational Notes:

Allow adding notes during usage, such as:

* Faults (e.g., sudden issue)
* Damages (e.g., broken mirror)
* Fuel entries (with quantity)
* Any operational remarks

4️⃣ User Interfaces

👷 Admin Dashboard:

* View all vehicles
* View vehicle status:

  * Available
  * In use
* Manage assignments
* View full activity logs per vehicle
* Filtering & search (by employee, date, vehicle)

🛠️ Technician Interface:

A simplified interface (mobile/web) that allows:

* Viewing the currently assigned vehicle
* Adding:

  * Fuel entries (with quantity)
  * Fault reports
  * General notes

5️⃣ Odometer Tracking

Description:

Track distance traveled per assignment period.

Requirements:

* Input odometer readings at:

  * Assignment start
  * Assignment end

Calculation:

* Distance = End Odometer – Start Odometer

Purpose:

* Fuel consumption analysis
* Vehicle usage evaluation

6️⃣ Reports & Printing

Report Types:

1. Vehicle Report:

* All users who operated the vehicle
* Time periods
* Distance traveled
* Notes and logs

2. Employee Report:

* Vehicles used
* Total distance
* Activity history

3. Full Fleet Report:

* Complete activity across all vehicles
* Filters by:

  * Date
  * Employee
  * Vehicle

Requirements:

* Export as PDF
* Print-friendly layout

7️⃣ Technical Structure (Suggested)

Core Entities:

1. Vehicles

* Stores vehicle data

2. Assignments

* Links vehicles to employees
* Includes:

  * start_time
  * end_time
  * odometer_start
  * odometer_end

3. Activity Logs

* Stores all events:

  * fuel
  * maintenance
  * notes
  * assignment actions

⚠️ Key Considerations

* Ensure data integrity
* Prevent overlapping assignments
* Maintain full audit logging
* Design for scalability (e.g., future GPS integration)

✅ Executive Summary

The system should connect:

* **Vehicle (Asset)**
* **Employee (User)**
* **Assignment (Usage Period)**
* **Activity Logs**

Within a unified platform that enables:

* Full tracking
* Operational insights
* Advanced reporting